

Recruited brown adipose tissue as an antiobesity agent in humans

.

Brown adipose tissue (BAT) burns fat to produce heat when the body is exposed to cold and plays a role in energy metabolism. Using fluorodeoxyglucose-positron emission tomography and computed tomography, we previously reported that BAT decreases with age and thereby accelerates age-related accumulation of body fat in humans. Thus, the recruitment of BAT may be effective for body fat reduction. In this study, we examined the effects of repeated stimulation by cold and capsinoids (nonpungent capsaicin analogs) in healthy human subjects with low BAT activity. Acute cold exposure at 19°C for 2 hours increased energy expenditure (EE). Cold-induced increments of EE (CIT) strongly correlated with BAT activity independently of age and fat-free mass. Daily 2-hour cold exposure at 17°C for 6 weeks resulted in a parallel increase in BAT activity and CIT and a concomitant decrease in body fat mass. Changes in BAT activity and body fat mass were negatively correlated. Similarly, daily ingestion of capsinoids for 6 weeks increased CIT. These results demonstrate that human BAT can be recruited even in individuals with decreased BAT activity, thereby contributing to body fat reduction.